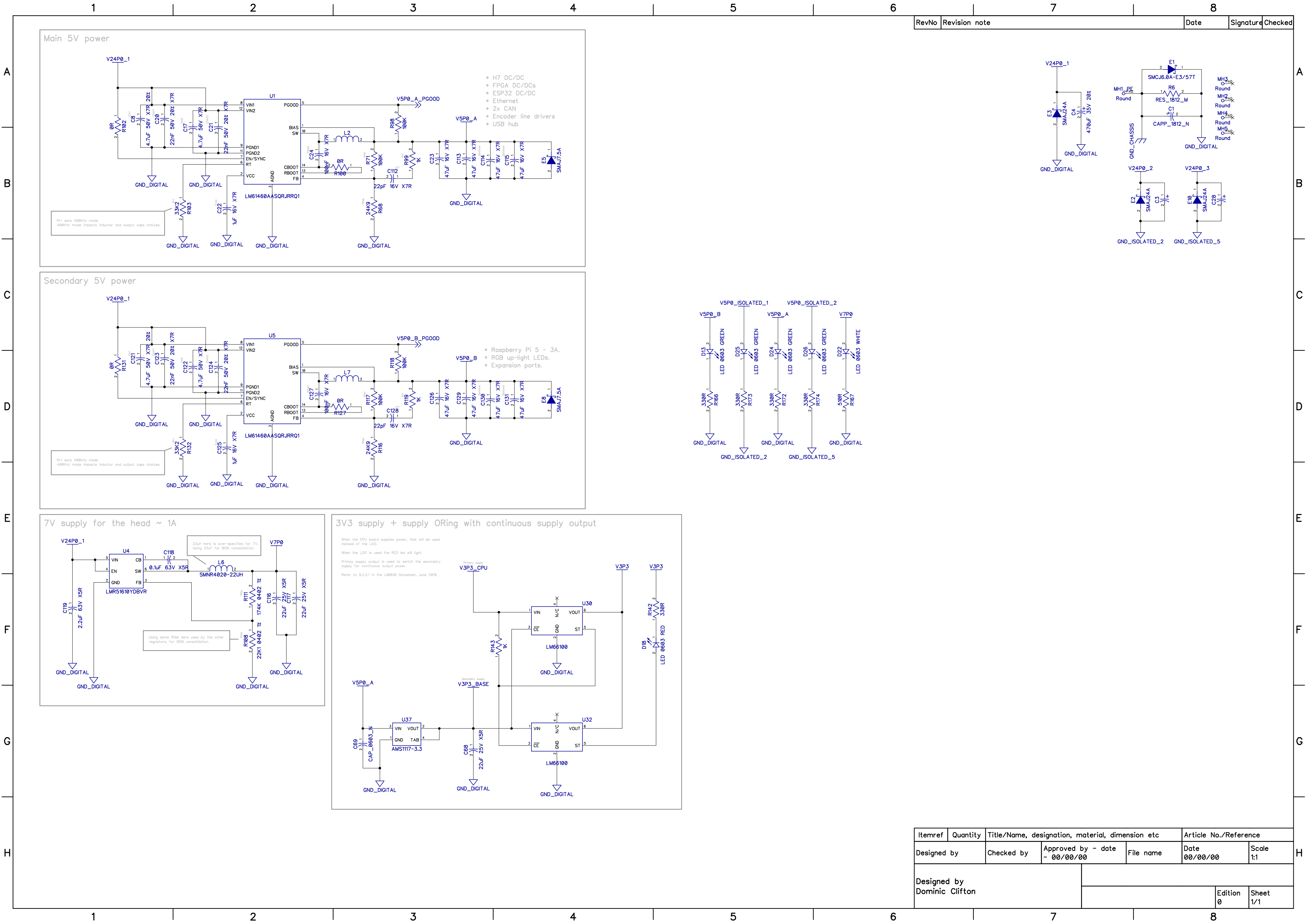
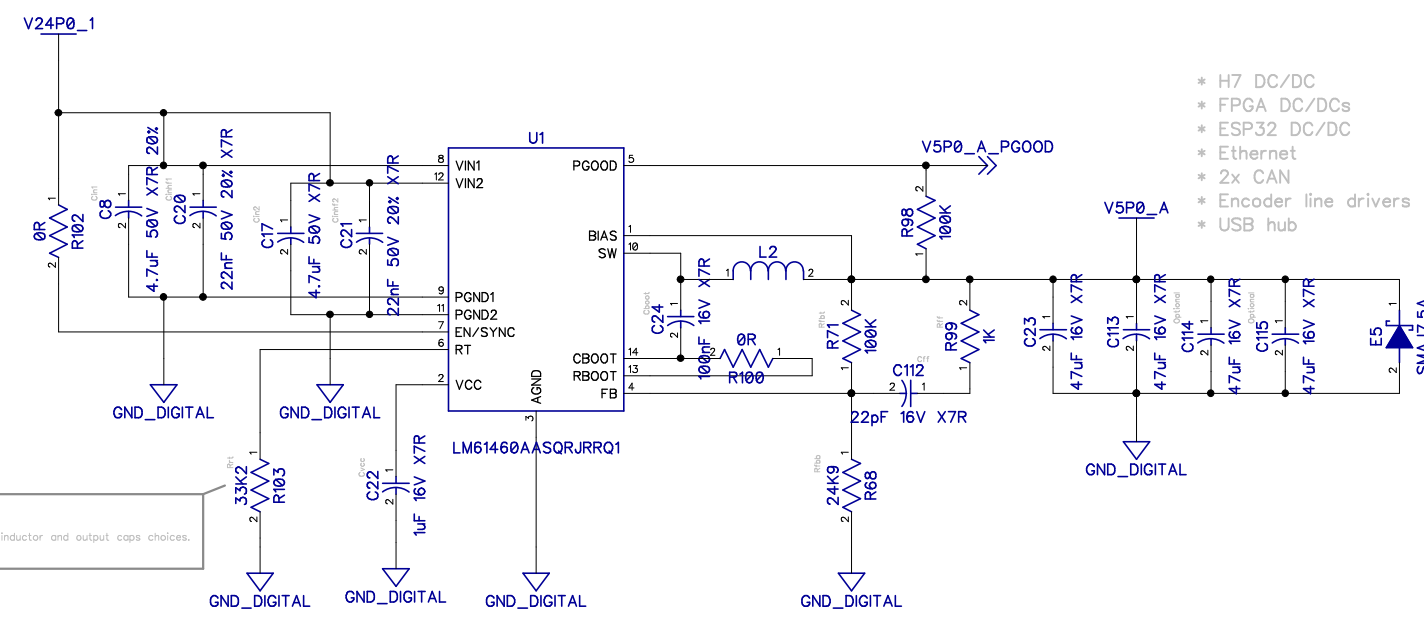


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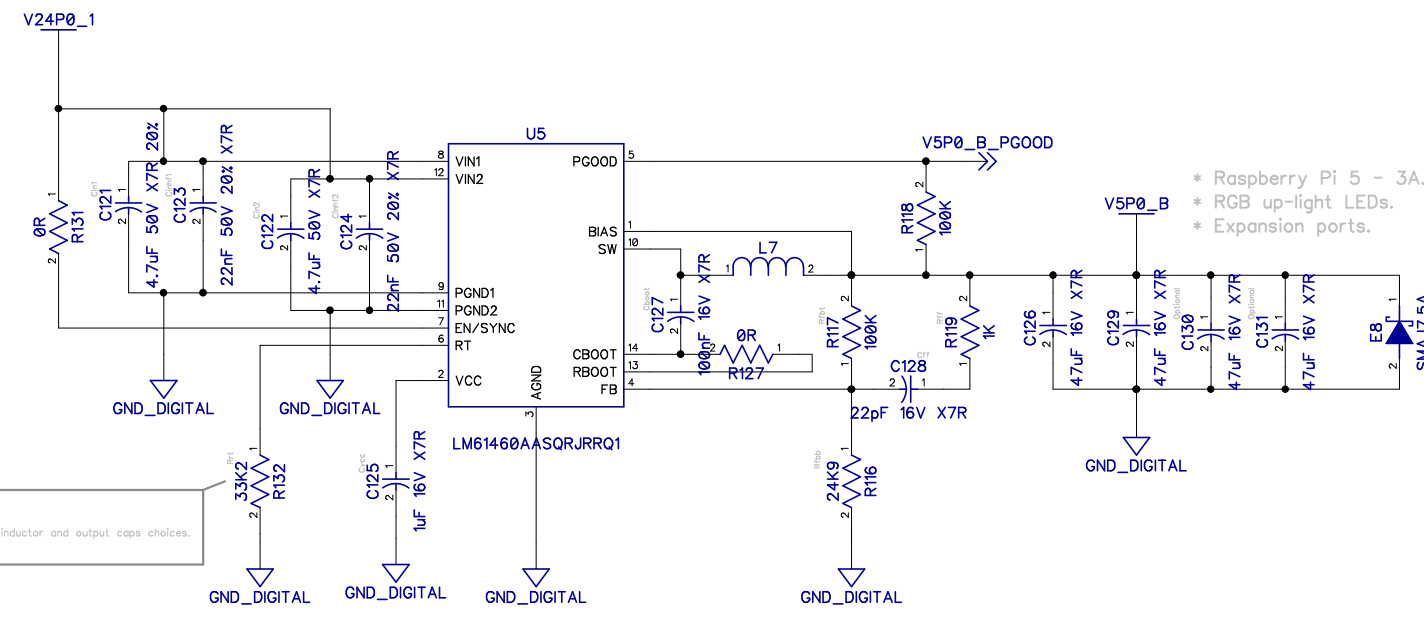


Main 5V power



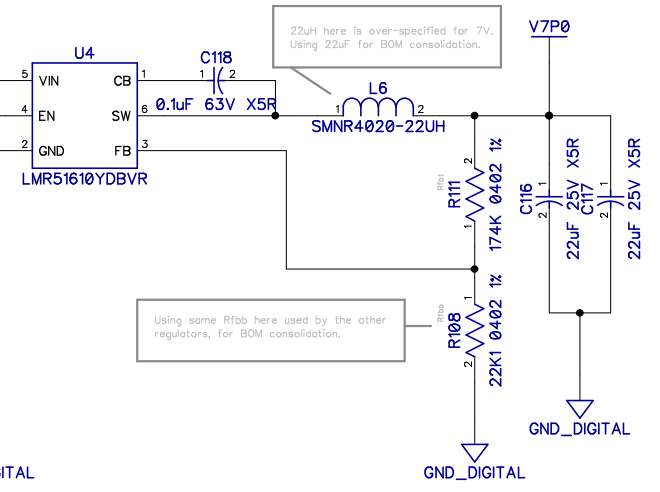
- * H7 DC/DC
- * FPGA DC/DCs
- * ESP32 DC/DC
- * Ethernet
- * 2x CAN
- * Encoder line drivers
- * USB hub

Secondary 5V power



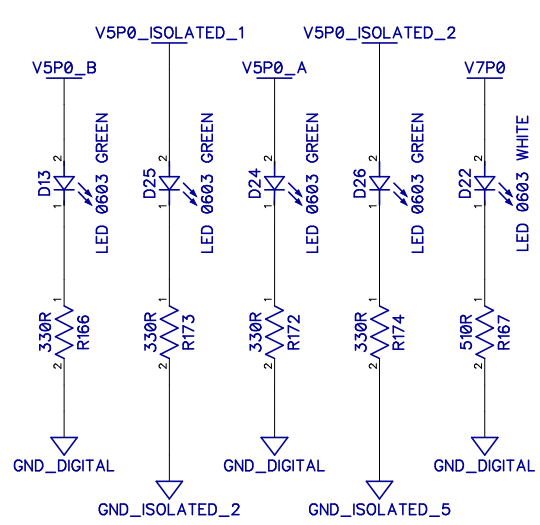
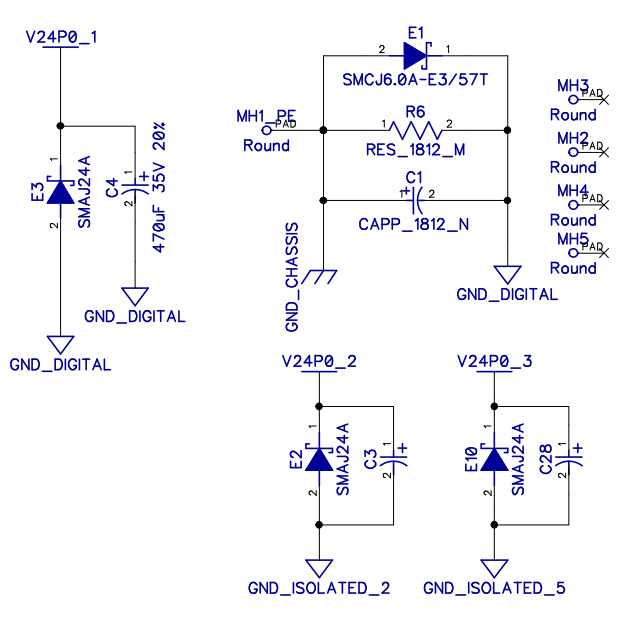
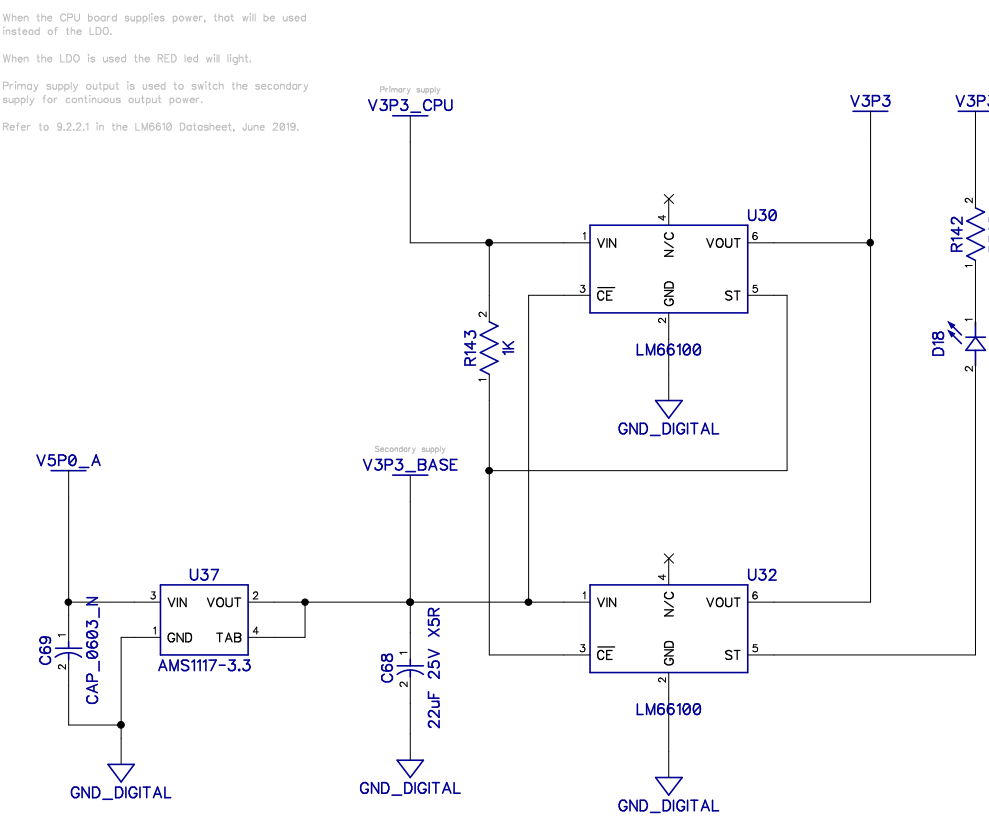
- * Raspberry Pi 5 - 3A.
- * RGB up-light LEDs.
- * Expansion ports.

7V supply for the head ~ 1A

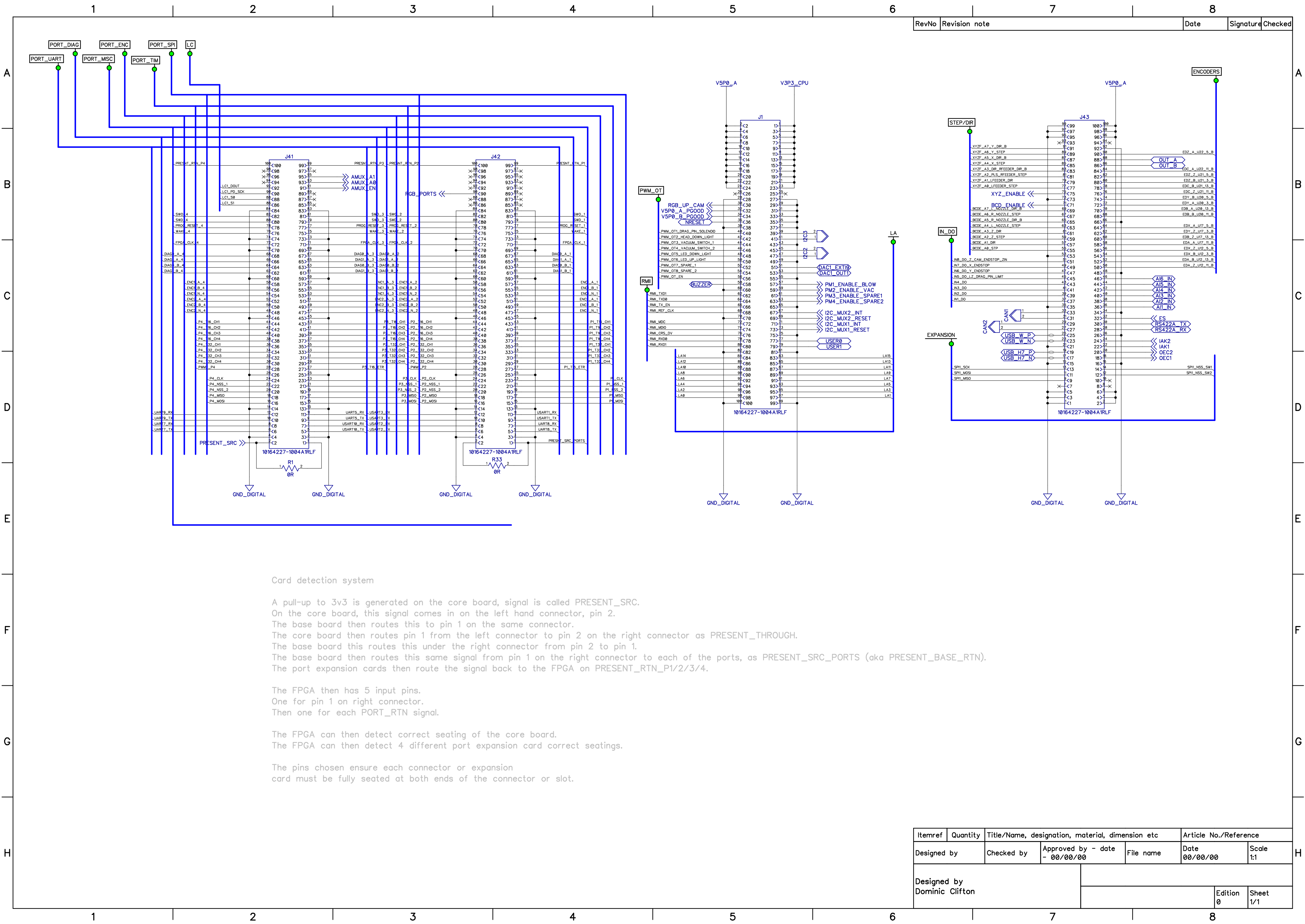


Using same R1b here used by the other regulators, for BOM consolidation.

3V3 supply + supply ORing with continuous supply output



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Card detection system

A pull-up to 3v3 is generated on the core board, signal is called PRESENT_SRC.
On the core board, this signal comes in on the left hand connector, pin 2.
The base board then routes this to pin 1 on the same connector.
The core board then routes pin 1 from the left connector to pin 2 on the right connector as PRESENT_THROUGH.
The base board this routes this under the right connector from pin 2 to pin 1.
The base board then routes this same signal from pin 1 on the right connector to each of the ports, as PRESENT_SRC_PORTS (aka PRESENT_BASE_RTN).
The port expansion cards then route the signal back to the FPGA on PRESENT_RTN_P1/2/3/4.

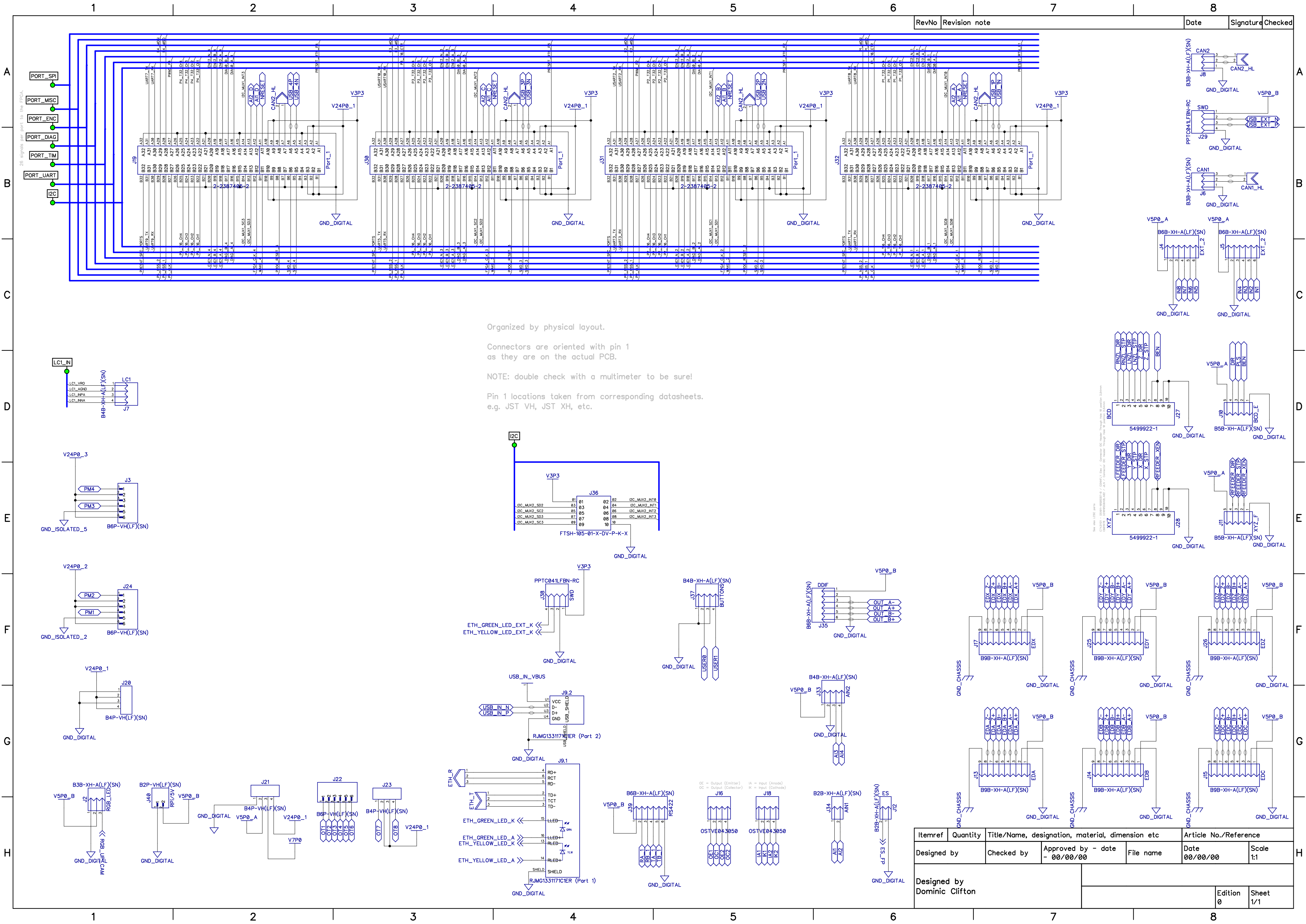
The FPGA then has 5 input pins.
One for pin 1 on right connector.
Then one for each PORT_RTN signal.

The FPGA can then detect correct seating of the core board.
The FPGA can then detect 4 different port expansion card correct seatings.

The pins chosen ensure each connector or expansion card must be fully seated at both ends of the connector or slot.

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H



Organized by physical layout.

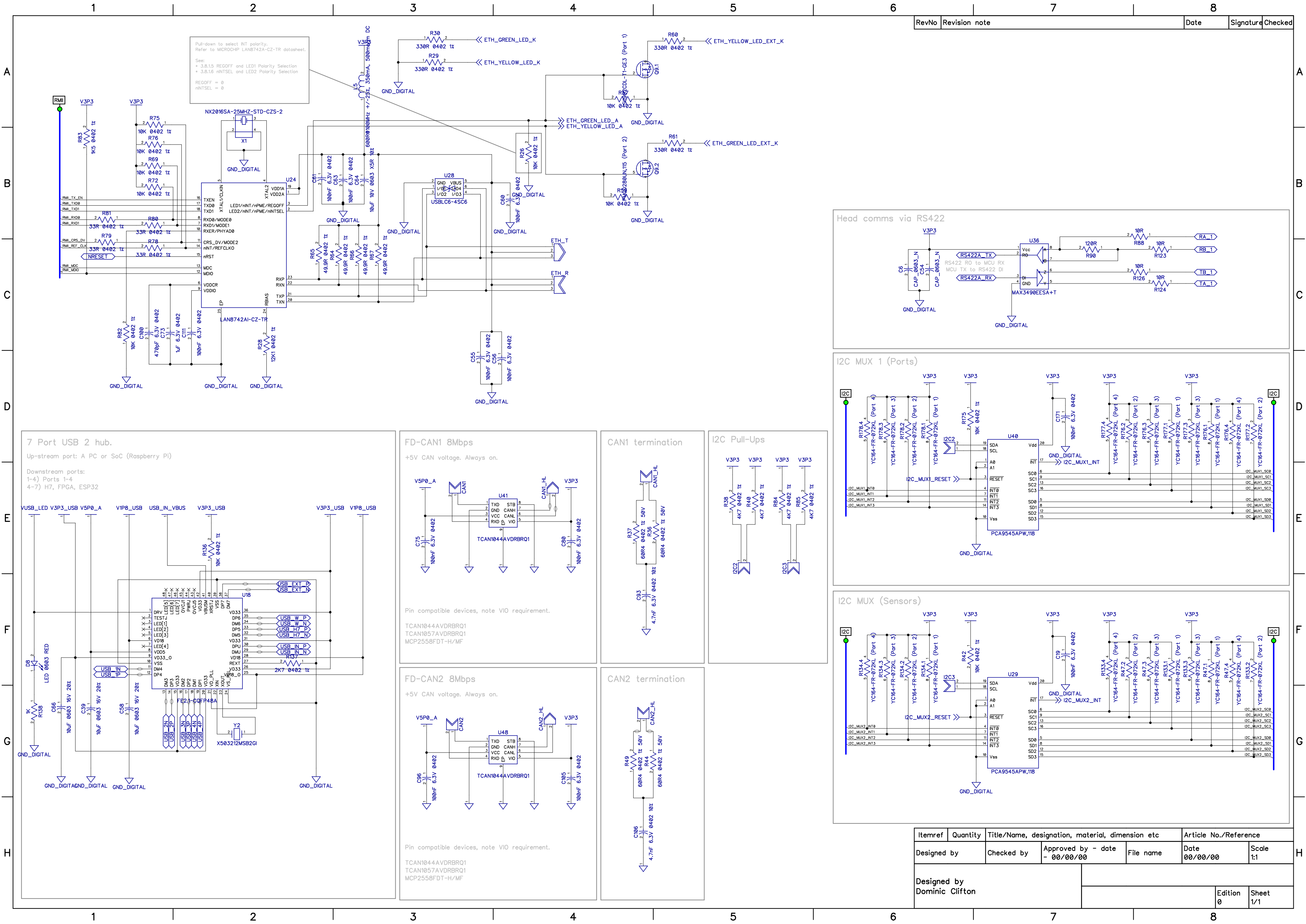
Connectors are oriented with pin 1 as they are on the actual PCB.

NOTE: double check with a multimeter to be sure!
Pin 1 locations taken from corresponding datasheets.
e.g. JST VH, JST XH, etc.

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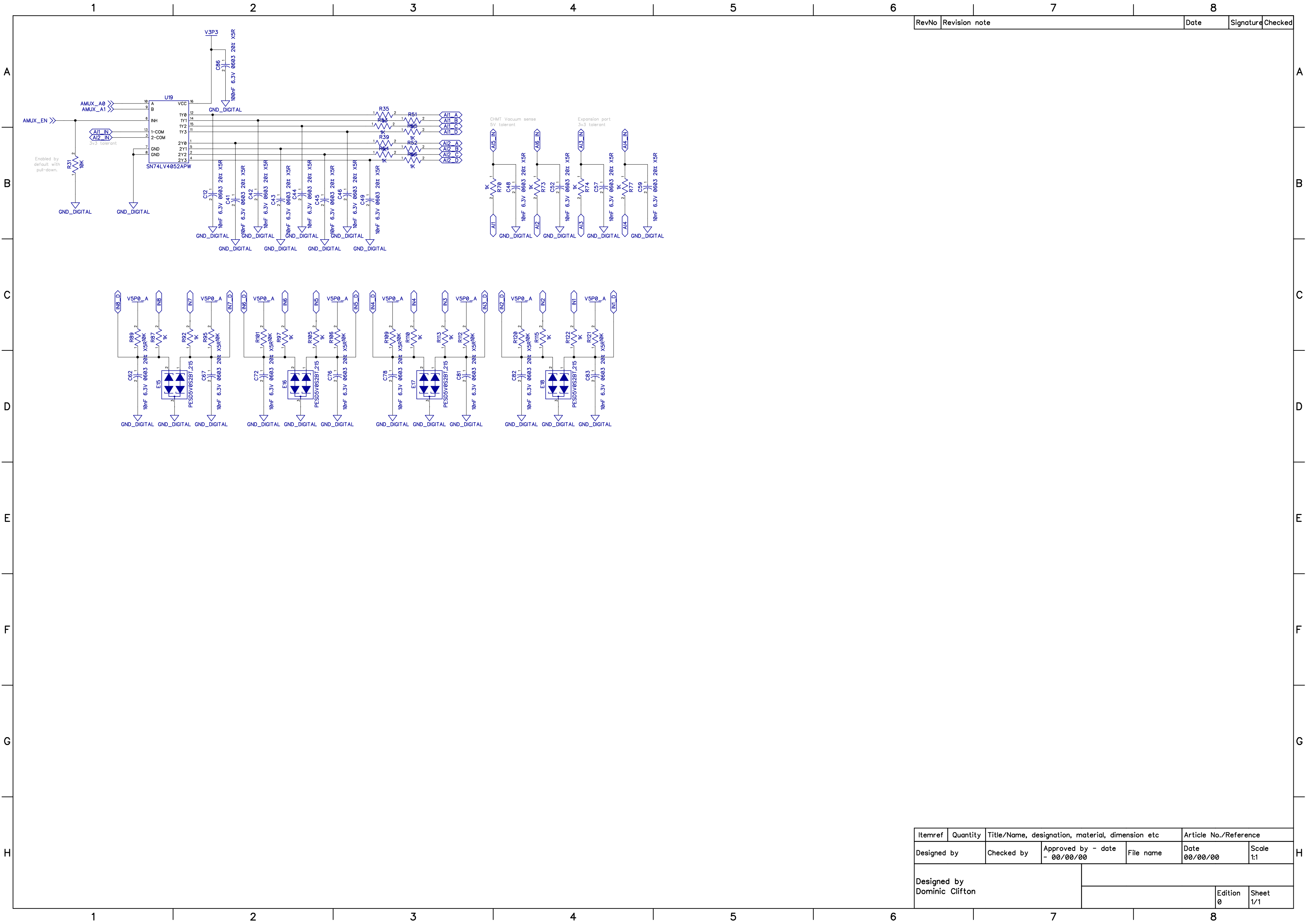
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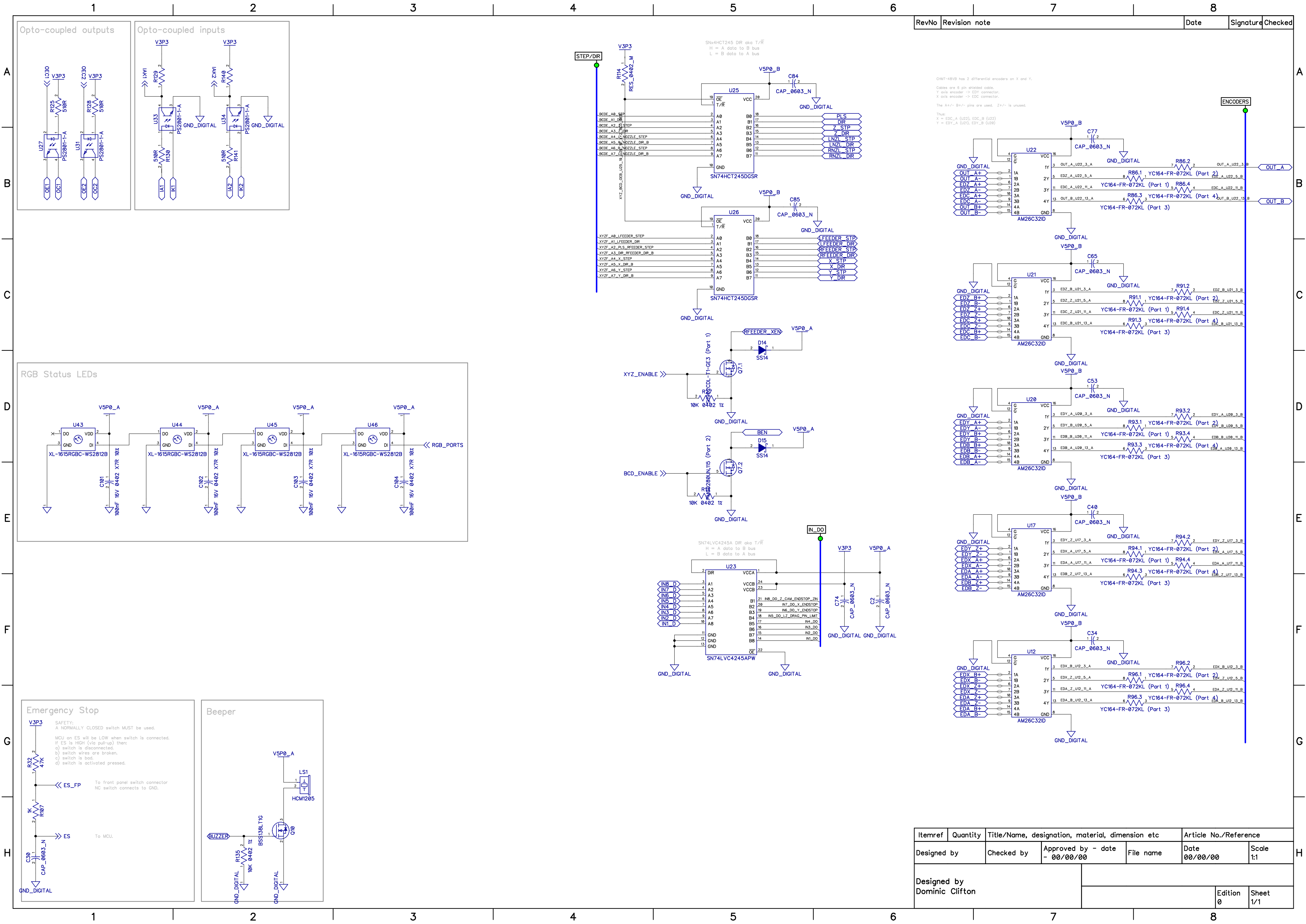
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GM7-48Vb has 2 differential encoders on X and Y.
Cables are 6 pin shielded cables.
Y axis encoder -> EDC connector.
X axis encoder -> EDC connector.
The A+/- B+/- pins are used. Z+/- is unused.

Thus:
X = EDC_A (U22), EDC_B (U23)
Y = EDC_Y (U21), EDC_Z (U26)

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